

Problem Set 2: Viral entry

Physics 230A/146A — Biological Physics

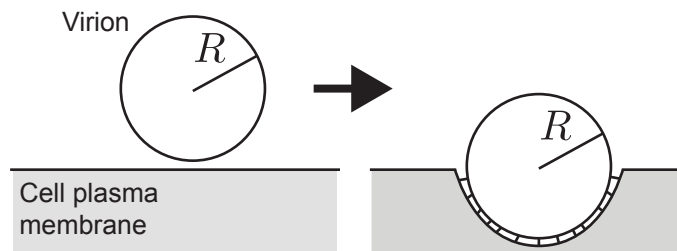
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Membranes, including the plasma membrane that constitutes the outer envelope of eukaryotic cells, are two-dimensional sheets composed of lipid bilayers that resist mechanical deformation, like bending. The bending energy of a surface can be modeled as

$$E = \iint \frac{1}{2} B \kappa^2 dA$$

where $B = 100 \text{ pN nm}$ is the bending modulus (note it has units of pN nm , unlike for rods and filaments), and κ is the mean curvature, and the integral $\iint dA$ is over the membrane's surface. For a sphere, $\kappa = 2/R$ where R is the radius of the sphere. For a cylinder, $\kappa = 1/R$ where R is the radius of the cylinder.

Viruses spread between cells as virus particles, or virions. These virions are not able to passively enter their host cells; in many cases, they must hijack the cell's endocytic machinery to help them actively gain entry.



- i. Suppose a viral capsid is approximately spherical with radius $R = 200 \text{ nm}$. How much energy is required to bend the plasma membrane to get the capsid half-way into the host cell?
- ii. Suppose the cell is under turgor pressure of $P = 0.01 \text{ pN/nm}^2$. Now, locally changing the volume of the cytoplasm results in an increase in energy $\Delta E = PV$, in addition to the energy required to bend the membrane. How much energy (bending plus volume change) is required to get the capsid half-way into the host cell?